

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular and Supplementary Summer 2024

Course: B. Tech.

Semester : IV

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Name & Code: BTEEC403 Electrical Machine- II

Max Marks: 60

Date: 18/06/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain construction and principle of operation of synchronous generator and motor	Remember	6
B) What is RMF? Find the Direction of RMF by using phasor method.	Understand	6
C) Enlist the causes of harmonics and explain How do you reduce the harmonics in EMF generated in alternator?	Understand	6
Q.2 Solve Any Two of the following.		12
A) Define the following terms with diagram and Formulas [2M each] 1. Coil Span Factor 2. Distribution Factor 3. Winding factor	Remember	6
B) What is Armature Reaction in Synchronous Machine? explain Armature Reaction at different power factors with Phasor Diagram	Remember	6
C) A 3-phase, 16-pole synchronous generator has a resultant air gap flux of 0.06 Wb per pole. The flux is distributed sinusoidal over the pole. The stator has 2 slots per pole per phase and 4 conductors per slot are accommodated in two layers. The coil span is 150° electrical. Calculate the phase and line induced voltages when the machine runs at 375 r.p.m	Evaluate	6
Q. 3 Solve Any Two of the following.		12
A) Draw the equivalent circuit diagram of synchronous machine and explain each parameter used in circuit.	Understand	6
B) Briefly discuss on synchronous condenser.	Understand	6
C) A 1500kVA, star connected, 2300V, 3-phase, salient-pole synchronous generator has reactance $X_d = 1.95 \text{ ohm}$ and $X_q = 1.4 \text{ ohm}$ per phase. All losses may be neglected. Find the excitation voltage for operation at rated kVA and power factor of 0.85 lagging	Evaluate	6

Q.4	Solve Any Two of the following.		12
A)	Compare squirrel cage and slipring Induction motor	Remember	6
B)	Derive the generalized torque equation of three phase induction motor and Obtain the condition for maximum torque under running condition	Create	6
C)	If a 3 phase 440V, 50Hz Induction motor is having 4 poles and operated at no load condition. Determine the rotor frequency of the motor.	Evaluate	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the construction and working principle of single phase Induction Motor	Understand	6
B)	Explain the double revolving field theory	Understand	6
C)	Briefly discuss on permanent magnet ac motors.	Remember	6

***** End *****